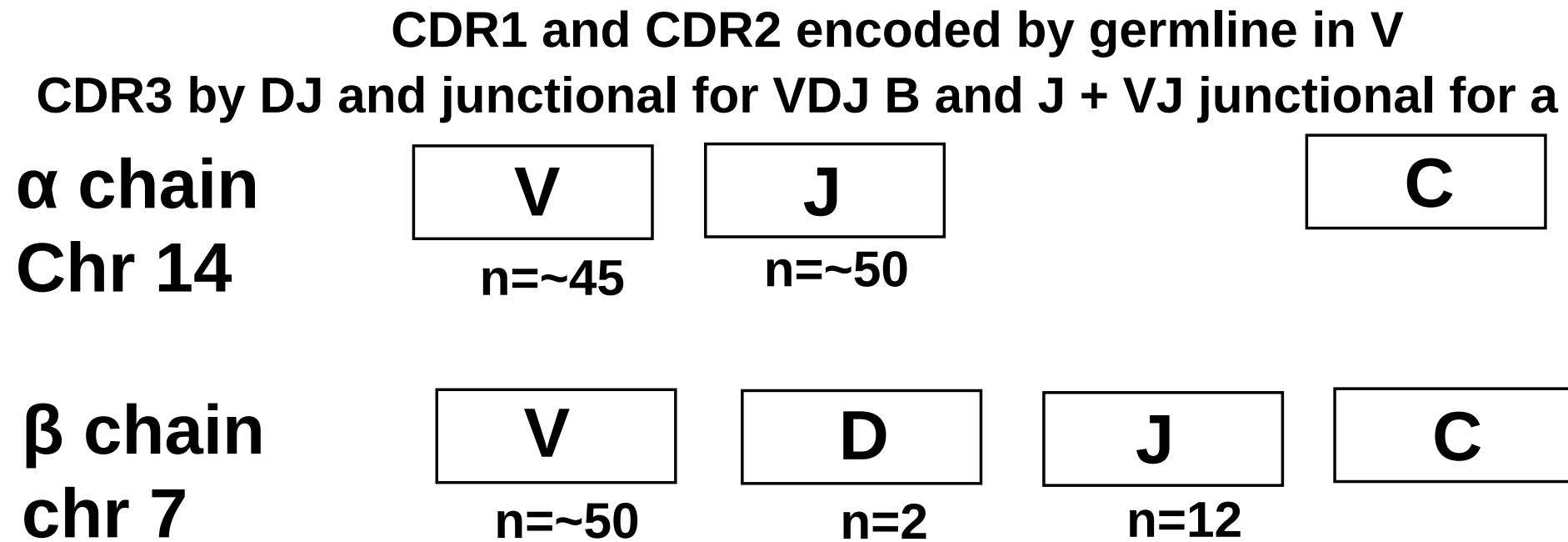




junctional diversity  
(non-germline)

largest contributor to diversity

P nucleotides (template after hairpin opening by Artemis)

N nucleotides (then nontemplate followed by DNA ligation) (20 by TdT) - more common in B chain

combinatorial diversity  
(germline)

1 to 3 million upper limit)

$10^{15}$  -  $10^{20}$  unique TCR combinations